

Polynomials, Piecewise Fits and the Road to Splines

For forty-one chapters the mean of the response has been a linear function of columns chosen in advance. This part keeps least squares, the projection geometry and every distribution theory built on them, and gives up one thing only: the insistence that the columns be the regressors themselves. The oldest way to do that is to let them be powers of a regressor, which makes a polynomial regression a linear model in a transformed basis, so that Parts II and III apply word for word — and that is what makes its failures instructive, because they are failures of the *family of shapes*, not of the fitting.

A polynomial is a global object: raising its degree to capture a feature at one end of the data changes the curve everywhere, sends the variance soaring at the edges, and turns any extrapolation into fiction. Worse, the obvious basis $1, x, x^2, \dots$ is one of the worst-conditioned matrices in numerical analysis. This chapter separates the two complaints. Orthogonal polynomials repair the arithmetic completely and the statistics not at all; knots repair the statistics, and lead to splines.

What you need. [Chapter 6](#) (projection, reparameterization, leverage), [Chapter 9](#) (sequential sums of squares, Example (A trend in the Nile)), [Section 1.8](#) (the condition number, Definition 1.8.1) and [Chapter 10](#) (what conditioning does to a fit), [Chapter 11](#) (restricted least squares), [Chapter 12](#) (bands) and [Chapter 14](#) (lack of fit). The later sections borrow the vocabulary of [Chapter 29](#) without needing its theory, and the closing discussion uses the bias–variance trade-off of [Chapter 19](#), the diagnostics of [Chapter 20](#), the transformations of [Chapter 22](#) and the causal reading of [Chapter 25](#). [Section 42.4](#) points forward to the penalized spline of [Section 43.3](#); that is a promise, not a prerequisite, and nothing here is proved from it. Of the chapters listed in the closing discussion, [Chapter 20](#) is a pointer only: it is named for the diagnostics a fitted spline should be put through, and nothing in this chapter cites it.

42.-1.1 Roadmap

- [Polynomial regression and its instability](#)
- [Orthogonal polynomials](#)
- [Piecewise polynomials](#)
- [Regression splines and truncated power bases](#)
- [Summary and notes](#)